Prepublication review & errata

17 October 2026

Dated correction and computational review

> **Read these results and data with caution.**
> The research and this review were generated by AI. They have not yet been independently verified by human experts or formalized in Lean. Computational success does not certify the categorical arguments on which a checker depends.

A companion to the historical manuscripts

R8_2_24 is reopened. The accepted continuation now covers 35 of 38 original cases. R6_4_155, R6_8_1660 and R8_2_24 remain open. This dated correction takes precedence over older claims of complete rank-eight classification through multiplicity two.

Audited research baseline: `e6a36a6f247b`. The online review carries subsequent reconciliation and the precise scope of each replay.

Rank three: induced-unit identity

In Proposition 6.3, “A uniform near-group exclusion” (p. 10), the sentence immediately before equation (6.2) must use **three** copies of the unit:

$$(\mathrm{F} \circ \mathrm{I})(1) = 3{\cdot}1 \oplus \mathrm{Y} \oplus \mathrm{mX}$$

Indeed, $1^2 + Y^2 + X^2 = 1 + 1 + (1 + Y + mX)$. The following restrictions of the central summands already have the correct total. This typographical correction does not change the argument or its conclusion.

In the same proposition, the Ng-Schauenburg formula cited as being “in Section 3” should refer to **equation (4.2), Section 4** (source label `eq:moments`).

Current repository review | Public review page

CORRECTED PROOF DEPENDENCIES

Rank six

Two passages cite Proposition 3.27, whose scope is R6_3_215, for reductions concerning different rings. Use the following dependencies in the frozen manuscript. The reference corrections do not settle either residual existence question.

1. Theorem 11.7, proof on p. 155

At the common-field Galois conjugation step, read:

> Extend to this common field the Galois conjugation supplied for **R6_4_155** by **Proposition 7.1**, and for **R6_8_1660** by the unrestricted sphericalization reduction at the beginning of **Section 7.5**, certified in the unrestricted reduction file below.

Source: `frontier_tensor_functors.tex`, lines 105-106. Correct labels: `prop:nonselfdual-neargroup-spherical` and `subsec:frontier-1660`.

2. Corollary 12.2, proof on p. 157

At the reduction preceding the full twist-one support argument, read:

> Apply the unrestricted sphericalization reduction for **R6_8_1660** described in **Section 7.5** (the same certificate), followed by **Lemma 11.6** on pseudo-unitary tensor functors.

Source: `dimension_block_20261016.tex`, lines 81-82. Lemma label: `lem:pu-tensor-functors`. Both chapter paths are relative to `manuscripts/rank6/papers/chapters/`.

Unrestricted reduction certificate

> `supplements/rank6/research/round4/m8/R6_8_1660_reduction.json`

The audit freshly replayed the unrestricted reduction: 640 signed quotients, 24 sign orbits and 58 nongauge character rejections. The retained original gauge orbit supplies the spherical section. This is a reduction check, not an existence or nonexistence proof for the source ring.

> **Both remain unresolved:** R6_4_155 and R6_8_1660. The rank-37 Verlinde obstruction rejects a particular witness; it does not exclude R6_8_1660 itself.

What the checks establish

R8_2_24 reopened: the categorical bridge is unproved

The projective-inner-square argument claims a decorated-phase cancellation with an incorrect incoming index. The condition i=k does not make the two phases cancel without an additional relation. Its two diagonal D flips and resulting six-slot reduction are therefore unjustified.

A replacement retaining all eight even D slots gives the required scalar consequences. Its unit normalization and finite algebra have been checked, but four mixed B/C categorical transports still lack a proof accounting for the oriented crossing and skew g self-duality. Two separate AI reviews did not certify that step. The conditional repair contributes no classification decision.

R8_2_24 is reopened. The accepted continuation now covers 35 of the original 38 cases: 28 proposed exclusions and seven unitary constructions. R6_4_155, R6_8_1660 and R8_2_24 remain open. Rank eight records 895 exclusions, 77 models and one open case through multiplicity two; its complete contiguous range ends at multiplicity one. Older complete-through-two claims are superseded by this correction.

REPRODUCIBLE CHECKS

Current repository replay

From the repository root, restore the split data and include historical stages explicitly:

```
python tools/restore_data.py
python tools/verify_review_20261017.py --historical
python -O tools/verify_review_20261017.py --historical
```

Without `--historical`, the dispatcher omits historical constructor replays. Original checkers that use `assert` must run in normal Python: an optimized invocation alone removes those checks.

The portable fermion checker replaces the scratch-only command in the frozen rank-six research note:

```
python evidence/unified_20261016/rank6/verify_fermion_crossing.py --repo .
```

Separate source identity, calculation and proof

The full rank-seven and rank-eight base-release loops passed in normal Python. Later transitions and the corrected current record passed in normal and optimized Python with exact parent restoration and source bindings. Historical arithmetic success does not certify the withdrawn R24 categorical step.

The reconstructed rank-five package passed all 40,999 pinned classes, 17 model matches and 40,982 general exclusions in both modes. Its new ledgers agree byte-for-byte. The original complete ledger was not recovered; its older hash remains a historical reference only.

Both complete rank-four replays passed: 889,530 rings, 15 models and 889,515 exclusions. Each regenerated 6,470,006 signed orbits; an independent full unsigned recount also passed.

The rank-six campaign covers all 25,362 exclusion certificates: 25,355 passed initially and seven passed complete unchanged checks under the Python backend after equation-ordering failures. The original failed run is retained. Both checks skipped after its failure gate passed separately; a hash-bound aggregate verified complete coverage. This is a combined campaign, not a successful single full run. The new portable orchestrator has not itself been run end to end.

Read the live reconciliation record

docs/PREPUBLICATION_REVIEW.md
GitHub Pages / review.html

The linked record takes precedence over this dated snapshot for later review outcomes. Source integrity, exact finite replay, categorical proof validity and independent expert review are distinct claims.